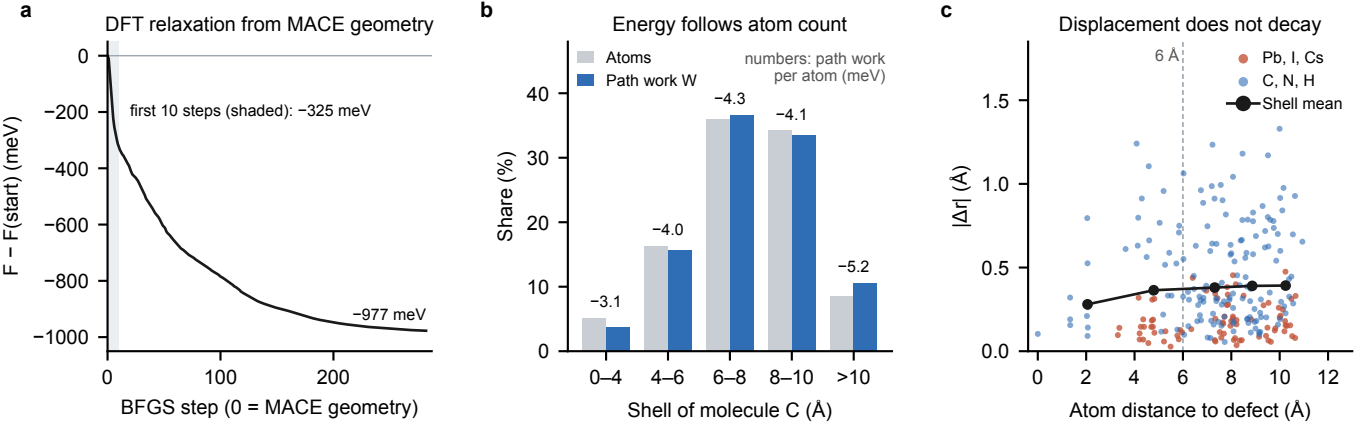
a
DFT relaxation from MACE geometry
first 10 steps (shaded): −325 meV
−977 meV
F − F(start) (meV)
BFGS step (0 = MACE geometry)
b
Energy follows atom count
Atoms
Path work W
numbers: path work per atom (meV)
−3.1
−4.0
−4.3
−4.1
−5.2
Share (%)
0–4
4–6
6–8
8–10
>10
Shell of molecule C (Å)
c
Displacement does not decay
6 Å
Pb, I, Cs
C, N, H
Shell mean
|Δr| (Å)
Atom distance to defect (Å)